

RUDY GAILLOT

Apprentice Embedded Electronic Systems Engineer

📍 Villemoyenne (Aube, France) 📞 +33 6 66 39 31 82 ✉️ rudy.gaillot10@gmail.com
🚗 Driver's license + personal vehicle 📅 Oct 14, 2005

Admitted to ISTP / Mines Saint-Étienne for a work-study engineering program, starting Fall 2026. Rotation schedule: 3 weeks corporate / 4 weeks academic (1st year), then 4/3 and 5/2.



PROFILE

Student in a 3-year Bachelor of Technology (BUT) in Electrical Engineering and Industrial Computing (GEII, specializing in ESE) at the IUT of Troyes. Admitted to the Embedded Electronic Systems Engineering degree (RNCP 40419) at ISTP / Mines Saint-Étienne under a work-study scheme. Experienced in industrial R&D/design offices, software/HMI development, embedded systems, and electronic prototyping.

TECHNICAL SKILLS

- **Embedded Systems & Programming:** C/C++, VHDL on Intel/Altera Cyclone FPGA, Linux, Git/GitHub, basic knowledge of Python, Java, and HTML.
Targets: AVR, ESP32, Raspberry Pi, notions of STM32/ARM.
Communication Buses: I2C, SPI, UART, CAN.
- **Electronics & Hardware Design:** KiCad (schematic capture PCB routing), LTspice, analog and digital electronics, op-amps, filtering, signal conditioning, basic knowledge of EMC and power electronics.
- **Automation & Industrial Computing:** Unity Pro (Ladder, FBD), PID regulation, control systems, real-time data acquisition and processing, FFT/Fourier, networking basics.
- **Testing, Measurement & Validation:** Oscilloscopes, logic analyzers, function generators (GBF), multimeters, soldering/prototyping, troubleshooting and testing of electronic boards, French electrical safety certifications (B1V and VAT).
- **Tools & Technical Environments:** Quartus (VHDL/FPGA), Qt Creator, Arduino IDE, Visual Studio, MATLAB, LabVIEW.
- **Engineering Methodology:** Functional analysis, technical specifications (cahiers des charges), datasheet interpretation, technical documentation, manufacturing files, and technical reports.

PROFESSIONAL EXPERIENCE

AUBE ÉLECTRONIQUE – R&D / Design Office

Saint-Thibault (10), France

R&D Engineering Technician Intern (3rd year BUT – 14 weeks)

March 2026 – June 2026

- Software development for an agri-food industry machine: HMI, control algorithms, and client-specific projects.

R&D Engineering Technician (Short-Term Contract)

April 2025 & June – July 2025

- Industrial software development: HMI programming and machine operational logic.

R&D Engineering Technician Intern (2nd year BUT – 8 weeks)

Jan. 2025 – March 2025

- Development setup for an agri-food industry machine: drafting specifications, component selection, and Raspberry Pi programming.

GRAPE HARVESTING – Éric Prignot

2023 – 2025

- Seasonal Vineyard Worker – Grape harvesting in Les Riceys (Aube, France).

ACADEMIC & PERSONAL PROJECTS

- **FPGA & Vision:** Camera configuration on Intel/Altera Cyclone FPGA using VHDL with RAM reading in C++ for image processing.
- **Image Processing:** Shape recognition and computer vision using Python.
- **Robotics:** Autonomous obstacle-avoiding robot (sensors, motorization, embedded logic).
- **Motor Control:** Motor control via FPGA and embedded control logic.
- **IoT / ESP32:** Server room temperature regulation via automatic control of 230V fans using an ESP32, sensors, and Home Assistant.
- **Control Systems:** Temperature regulation loop using sensors and electronic feedback loop.
- **Energy:** Photovoltaic systems connection, MPPT concepts, and EV charging stations.
- **Infrastructure:** Linux environment installation and network configuration with router/switch.

EDUCATION

2023 – 2026

IUT de Troyes (France)

Bachelor of Technology (BUT) in Electrical Engineering and Industrial Computing (GEII) – ESE Pathway

2020 – 2023

Lycée Chrestien de Troyes (France)

French High School Diploma (Baccalauréat Général)

Specialties: Mathematics, Physics-Chemistry, and Advanced Mathematics (Maths Expertes)

LANGUAGES

French: Native **English:** B2 (Upper-Intermediate) **Spanish:** A2+ (Elementary)